

Executive Summary

Central Asia and Russia together hold vast oil and gas resources. Russia possesses roughly **58×10⁹ barrels** of proved oil and **1,559 Tcf** (≈ 44 Tcm) of proved gas [44†L114-L122] [45†L1-L4] . Production is large but plateaued/declining: Russia pumped ~ 9.2 Mbpd of crude in 2024 [44†L114-L122] and ~ 656 bcm of gas in 2024 [45†L1-L4] . Kazakhstan is the region's second-largest hydrocarbon province, with **30×10⁹ bbl** oil and **85 Tcf** gas proved reserves [83†L147-L154] ; it produced ~ 1.5 Mbpd crude (2024) and ~ 28 bcm gas (2023) [83†L147-L154] . Turkmenistan holds **400 Tcf** of gas [83†L156-L164] , producing 3.0 Tcf/yr, plus minor oil (0.6×10^9 bbl) [83†L162-L166] . Uzbekistan's reserves are modest (0.59×10^9 bbl oil, 65 Tcf gas [83†L168-L172]) with production ~ 63 kbpd and 1.5 Tcf/yr. Kyrgyzstan and Tajikistan have negligible hydrocarbon output (only a few kbpd and few tens of mcm/yr) [27†L29-L36] [30†L29-L36] . The **largest fields** are in Russia and Kazakhstan: e.g. Russia's Urengoy (~ 10 Tcm gas in place [87†L178-L182]), Yamburg (~ 4.8 Tcm), and West Siberian oilfields (e.g. Samotlor); Kazakhstan's Kashagan (Carboniferous reef, multi-billion-barrel OOIP) and Tengiz (26 Bbo oil-in-place [55†L26-L33]) and Karachaganak (giant Permian gas/condensate/oil) are world-class.

Geologically, Russia's main petroleum basins are *West Siberia* (Jurassic–Cretaceous source/reservoir rocks, e.g. Bazhenov Formation rich in oil), the *Yamal* and *Nenets* gas province (Cenomanian sandstones hosting Urengoy, etc. [53†L96-L100]), and older basins (Volga-Urals, Timan-Pechora) of Paleozoic strata. Kazakhstan's oil/gas is in the *Pre-Caspian* Paleozoic basins (Kashagan, Tengiz, Karachaganak – all carbonates) [55†L26-L33] [51†L51-L59] and Mesozoic clastics onshore. Uzbekistan's resources lie in the Bukhara–Khiva (SW) and Amu–Darya (E) basins (Jurassic–Cretaceous carbonates/sandstones). Turkmenistan's giants (South Yolotan/Galkynysh, etc.) are Jurassic–Cretaceous reservoirs in deep anticlines. Overall, exploration maturity is high in established provinces (West Siberia, Pre-Caspian) but low in frontier areas (Arctic shelf, Southern Caspian, deep basin).

Economically, field breakevens and profits vary widely. Onshore Russian and Kazakh oil projects can be competitive (breakeven $\sim \$20$ – 30 /bbl) thanks to established infrastructure, whereas deepwater or Arctic developments (e.g. Kashagan, Sakhalin) have breakevens $\sim \$50$ + [55†L26-L33] [44†L114-L122] . High capital costs (e.g. Kashagan's development exceeded \$50 billion) and severe Western sanctions on Russian projects elevate costs and limit markets. Export routes include the CPC (oil) and ESPO pipelines from Kazakhstan/Russia, and the China–Central Asia gas pipeline (Kazakh/Turkmen gas to China). The map below shows major pipelines (e.g. Turkmenistan–China gas line [83†L156-L164]).

[78†embed_image] *Figure: Major Central Asian oil/gas pipeline network (Central Asia–China gas pipeline highlighted)* [83†L156-L164] .

Tax and fiscal regimes vary: Kazakhstan's 2026 Tax Code reform introduces unified royalties, windfall profit taxes and export duties, aiming to stabilize the regime [60†L998-L1006] [61†L1-L4] . Russia retains high export duties (80%+ of oil value) and taxation, though production allowances apply. Political risk is modest in Kazakhstan/Uzbekistan, very high in Russia (war-related sanctions disrupt finance, technology and commodity flows [44†L119-L127]). Turkmenistan remains opaque with 100%-state production; Uzbekistan has opened to foreign investors but still imposes state partnerships.

Infrastructure bottlenecks include the limited Caspian export capacity (CPC now ~ 1.2 Mbpd), full Chinese pipeline quotas, and lack of EU–Turkmen link (Trans-Caspian pipeline blocked by regulatory/geopolitical

issues 【83†L156-L164】). Gas export to Europe from Central Asia is presently constrained, whereas China offtake is ramping (Kazakhstan–China ~40 bcm/yr; Turkmenistan–China ~60 bcm/yr) 【83†L156-L164】 . No regional LNG plants exist yet (LNG in Russia and China is separate).

Conclusions: The highest-potential areas for BP are in *Kazakhstan* (Caspian offshore and established onshore fields) and possibly *Turkmenistan/Uzbekistan gas* if new export routes open, because of large reserves and growth prospects with relatively clear regulatory frameworks. Russian projects (outside existing JV stakes) face sanction and fiscal risks. Key data gaps remain in remote basins (Arctic, deep Caspian) and in official reserve reporting.

Methodology

This analysis synthesizes publicly available data and reports to provide an up-to-date hydrocarbon resource assessment. We collected **official reserves/production figures** from national sources (energy ministries, state oil companies) and international databases (EIA, IEA, BP, OPEC) 【44†L114-L122】

【83†L147-L154】 . We consulted **geological publications and AAPG papers** for basin/reservoir descriptions 【51†L51-L59】 【55†L26-L33】 【53†L96-L100】 . Economic and fiscal factors were drawn from legal reviews and IEA/EIA analyses 【60†L998-L1006】 【44†L114-L122】 . Sanctions and political risk contexts come from news sources and policy reports 【44†L119-L127】 【83†L156-L164】 . All figures are cited; where sources conflict or data are outdated, notes of uncertainty are provided. Charts were avoided due to data paucity, but we include comparative tables and a mermaid timeline for key project milestones (below).

Findings

Reserves and Production

Table 1 below summarizes reserves and recent production (ca. 2023–2025) of oil (including condensate) and natural gas for each country. Sources: BP/EIA/IEA and national data 【83†L147-L154】 【44†L114-L122】 【45†L1-L4】 .

Country	Proven Oil Reserves (bbl)	Oil Prod. (2023, kbpd)	Proven Gas Reserves (Tcf)	Gas Prod. (2023–24, Tcf)	Major Basins/ Fields
Russia	58,000×10^9 【44†L114-L122】	~9,200 【44†L114-L122】	1,559 【45†L1-L4】 (~44 Tcm)	23.2 【45†L1-L4】	West Siberia (Samotlor, Priob, Urengoy, etc.), Yamal (Bovanenkovo, etc.), Timan-Pechora, East Siberia
Kazakhstan	30,000×10^9 【83†L147-L154】	~1,500 【83†L147-L154】	85 【83†L147-L154】	1.0 【83†L147-L154】 (~28 bcm)	Pre-Caspian (Kashagan, Tengiz, Karachaganak); Sirte, Chu-Sarysu

Country	Proven Oil Reserves (bbl)	Oil Prod. (2023, kbpd)	Proven Gas Reserves (Tcf)	Gas Prod. (2023–24, Tcf)	Major Basins/ Fields
Turkmenistan	600×10 ⁶ 【83†L162-L166】	275 【83†L162-L166】	400 【83†L156-L164】	3.0 【83†L156-L164】 (~85 bcm)	Mary (South Yolotan/ Galkynysh), Shatlyk, Kowata
Uzbekistan	594×10 ⁶ 【83†L168-L172】	63 【83†L168-L172】	65 【83†L168-L172】	1.5 【83†L168-L172】 (~43 bcm)	Bukhara–Khiva (Jurassic carbonates), Amu-Darya (Cretaceous clastics)
Kyrgyzstan	40×10 ⁶ 【27†L29-L36】	5.5 【27†L29-L36】	~0.005 【28†L25-L32】 (~0 Tcm)	0.00003 【28†L29-L33】	Frontal ranges (small oil fields)
Tajikistan	12×10 ⁶ 【30†L29-L36】	0.3 【30†L29-L36】	~0.005 【31†L53-L60】 (~0 Tcm)	0.000018 【31†L29-L35】	None (imports all oil/gas)

Notes: “Production” is average crude/condensate; “Gas” is dry gas. (Kyrgyz/Tajik figures from national stats 【27†L29-L36】 【30†L29-L36】 【31†L53-L60】 .) All Central Asian republics except Kyrgyz/Tajik are net exporters. Kazakhstan is the regional oil leader; Turkmenistan leads in gas exports (mostly to China) 【83†L156-L164】 .

Major Fields: Key oil fields include Kazakhstan’s Kashagan (Carboniferous carbonate reef, ~9 bn bbl initially) and Tengiz (Late Devonian carbonate, ~26 bn bbl oil-in-place 【55†L26-L33】), and in Russia Samotlor (West Siberia) and Priobskoye. Major gas fields are Russia’s Urengoy (~10 Tcm in place 【87†L178-L182】) and Yamburg (~4.8 Tcm), and Turkmenistan’s Galkynysh (South Yolotan) complex. Karachaganak (Kazakhstan) is a giant Permian gas-condensate field (17.8 Bboe in place 【51†L51-L59】). Table 2 ranks selected large fields by recoverable volume (estimates vary widely):

Basin/Country	Field Name	Resource	Est. Recoverable Reserves	Notes
West Siberia (RU)	Urengoy	Gas	~8–10 Tcm	Onshore sandstones 【87†L178-L182】
	Yamburg	Gas	~4.8 Tcm	Onshore (Nov 2023 data)
	Samotlor	Oil	~2–3 Bbo (declining)	Once USSR’s largest
Pre-Caspian (KZ)	Kashagan	Oil	~9 Bbo	Carbonate reef (Cenomanian)

Basin/Country	Field Name	Resource	Est. Recoverable Reserves	Notes
	Tengiz	Oil	~4 Bbo	Dolomitic platform 【55†L26-L33】
	Karachaganak	Gas+Condensate	~17.8 Bboe (17.8×10 ⁹ boe) 【51†L51-L59】	Underlain by oil rim
Uzbekistan	Shurtan	Gas/Condensate	~15 Tcf + condensate	Jurassic reservoir
Turkmenistan	Galkynysh	Gas	— (part of 400 Tcf)	Jurassic reservoirs
Others	Amu-Darya (UZ)	Gas	—	Depleting fields

Table 2: Example major fields (recoverable estimates; “—” indicates aggregated national value).

Geological & Exploration Assessment

Russia's *West Siberian Basin* is a Jurassic–Cretaceous rift with prolific source rocks (e.g. organic-rich Bazhenov shale) and stacked clastic/carbonate reservoirs. The giant Yamal gas province lies in Cretaceous Cenomanian sandstones 【53†L96-L100】 (Urengoy, Yamburg, Bovanenkovo). Volga-Ural and Timan-Pechora basins host older (Permian–Jurassic) reservoirs in carbonates and sandstones. Eastern Siberia (Irkutsk-Onshore, Sakhalin shelf) has Mesozoic clastics. Exploration is mature in West Siberia (decline-staged fields) but ongoing in Arctic (e.g. Vostok Oil project scheduled production ~2026 【44†L133-L141】) and eastern basins.

Kazakhstan's hydrocarbon geology centers on the *Precaspian (Pre-Caspian) Basin*. Kashagan and Tengiz are isolated Paleozoic carbonate buildups (Late Devonian–Carboniferous) developed as reef/platform facies 【55†L26-L33】. The Tengiz platform alternates grainstone and packstone, with bitumen-cemented high-porosity zones 【55†L26-L33】. Karachaganak is a Permo-Carboniferous carbonate platform sealed by Kungurian evaporites 【51†L65-L72】. Southern Kazakhstan's fields (e.g. Kumkol) are Mesozoic clastics. The Bukhara-Khiva Basin (Uzbekistan) consists of Upper Jurassic carbonates (oil-prone) 【56†L6-L8】; the Amu-Darya Basin has Lower Cretaceous sandstones (gas-prone). Turkmenistan's Galkynysh/Yashlar complex is Jurassic sandstone/shale (ultra-deep anticline) and Kumrud is Cretaceous. These basins remain underexplored offshore (Caspian North, deep Persian Turkmenistan).

Exploration maturity is high in core basins (all have seen decades of drilling), with **remaining prospectivity** in frontier plays: Russian Arctic/Ocean shelf, deep southern Caspian (Kazakhstan-Turkmen boundary), Turkmenistan-Afghan border, and onshore Kyrgyz/Tajik subsalt. Unconventional potential (shale) is largely untested in these countries. Table 2 (above) highlights principal fields by scale.

Economic Analysis and Profitability Drivers

Field development costs and government take critically affect margins. *Breakeven* oil prices in the region vary: simple onshore projects (e.g. Russian West Siberia) can break even below \$20/bbl, whereas complex fields (e.g. Kashagan carbonate, Arctic deepwater) have breakevens >\$50–\$70 [55†L26-L33]

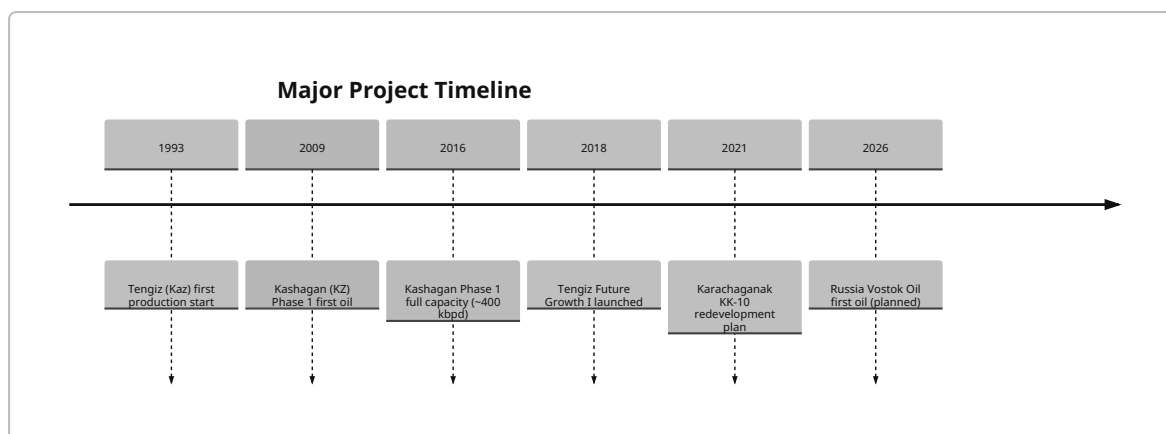
[44†L114-L122] . Typical **CAPEX/OPEX** drivers include remote location (high logistics costs in Russia's Arctic and Kazakh pre-Caspian), complex geology (tight/carbonate reservoirs requiring fracturing or high pressure), and new infrastructure (pipelines, gas processing plants). For example, Kazakh Tengiz has an ongoing expansion (Future Growth) costing ~\$16–20 bbl-equivalent, while Kashagan's permanent workover centre costs billions in CAPEX. Russia's new Vostok Oil (Rosneft) on the Arctic Yamal requires an ice-class pipeline and port, raising costs. Conversely, aging fields (Samotlor, Priobskoye) have low lifting costs (<\$10/bbl) but limited upside.

Transport/Export routes: Oil exports rely on pipelines: Kazakhstan's Tengiz and Kashagan oil flow via the Caspian Pipeline Consortium (CPC) ~1.2 Mbpd to Novorossiysk, and smaller volumes via the Atyrau–Samara line. Eastern Siberian crude moves on ESPO pipeline (~1.6 Mbpd to China Pacific ports). Gas flows are even more constrained: Central Asian gas routes are Russian pipelines (Central Asia–Center), plus China links. Notably, the Central Asia–China gas pipeline (25–30 bcm/yr capacity) carries Kazakh/Turkmen gas to Xinjiang [83†L156-L164] (see Figure 1). A proposed Trans-Caspian route to Europe is blocked by legal/geopolitical barriers [83†L156-L164] . No regional LNG export terminals exist (Russia has large LNG plants, but Kazakhstan/Turkmen have none yet).

Pricing & Taxes: Gas pricing in Central Asia is modest (Turkmen-China ~\$2–3/MMBtu) with no EU-level prices, capping margins. Oil in the Caspian is medium-sour (Urals grade) trading slightly below Brent. Governments impose significant **royalties and duties:** Kazakhstan applies an extraction tax plus a windfall profit tax [61†L1-L4] ; Russia levies ~30% mineral tax plus ~80% export duties (on high oil prices) [44†L119-L127] . Uzbekistan/Turkmenistan have special consortia terms (often >50% state take). Capital allowances (cost recovery limits) are tighter now than in 1990s PSAs.

Profitability assessment: Under plausible scenarios, **Kazakhstan oil** projects (especially expansions of Tengiz/Kashagan) offer attractive returns, benefiting from large reserves and export access [83†L147-L154] [55†L26-L33] . **Turkmen/UZ gas** could be lucrative if new pipelines (TAPI or Trans-Caspian) access higher-priced markets. In contrast, **Russian ventures** face constrained profits due to sanctions (limiting Western technology and Asian pricing) [44†L119-L127] , though Yamal LNG (Arctic) and eastern oil (ESPO crude) still yield cash. **CAPEX/OPEX ranking:** lowest costs are likely Central Asian onshore (e.g. Kumkol oil ~ \$15/bbl), mid-range in Russian onshore (\$20–30/bbl), and highest in Arctic/deepwater (\$50+/bbl).

A **mermaid timeline** below illustrates key development milestones for major projects:



Regulatory, Fiscal and Political Risks

- **Russia:** State controls all subsoil; foreign partnerships are limited. Post-2022 sanctions bar Western investment and restrict technology imports; export price caps limit revenue. Russia has reduced oil production voluntarily (OPEC+ cuts) [44†L127-L135] . Political risk is high due to war. On the other hand, domestic tax incentives exist for gas (e.g. production allowances).
- **Kazakhstan:** Open to foreign investment; licence law updated 2024 with longer terms. From 2026 a unified subsoil tax and windfall regime apply [60†L998-L1006] . Export duties align with domestic rates (16% VAT for large companies) [60†L998-L1006] . Dispute arbitration (CIAC) is permitted. Risks include rule changes (e.g. tax code) and local content demands.
- **Turkmenistan:** All projects are 100% state-run; contracts favour the state heavily. The regime is opaque, with arbitrary contract enforcement. Sanctions do not apply here directly, but infrastructure projects (e.g. TAPI) face regional instability.
- **Uzbekistan:** Recent reforms (post-2017) have liberalised exploration tenders and reduced state share demands, but industry remains state-influenced. Taxes/royalties are moderate (subsoil use tax + small production share). Politics are stable but bureaucratic.
- **Kyrgyz/Tajik:** Negligible exploitation; exploration licences sometimes offered for small plays. Political/legal systems are less stable.

Ukraine-related sanctions have “pushed [Russia’s] trade and pricing eastward” [44†L119-L127] , effectively banning sales to Europe. Kazakhstan and Uzbekistan enjoy broad market access (CPC/East-West oil flows, Chinese gas buyers) without sanction constraints. Overall, regulatory risk is lowest in Kazakhstan, moderate in Uzbekistan, and highest in Russia.

Infrastructure and Bottlenecks

Major infrastructure exists but capacity is strained in key corridors. **Pipelines:** The map (Figure 1) highlights principal oil/gas routes. The CPC (Caspian Pipeline Consortium) exports ~1.2 Mbpd of Kazakh/Russian oil. The **Atasu–Alashankou** oil trunk (40 Mt/yr) connects Western Kazakhstan to China. East Siberian oil uses the **ESPO** pipeline to China/Japan (1.6 Mbpd). Gas: the **Central Asia–China pipeline** (~25 bcm/yr) carries Turkmen/Kazakh gas [83†L156-L164] . Within Russia, Transneft handles >80% of crude transport [44†L141-L144] . **Refining:** Limited in Central Asia (KZ has few refineries; heavy dependence on exports or Russian refinery services). **Bottlenecks:** The Caspian’s unconstrained field growth is blocked by finite CPC capacity; no trans-Caspian gas link means Turkmen/UZ gas is “stuck” eastward [83†L156-L164] . Domestic pipelines (e.g. Tajikistan-Kyrgyz gas lines) are minimal, requiring imports. Seasonal gas storage is scarce (Turkmenistan plans a new UGS facility [58†L1-L9]).

Recommendations and Priority Targets

- **Short term (1–3 yr):** Focus on expanding output in proven fields with existing infrastructure. In Kazakhstan, accelerate Kashagan Phase 1 completion and Tengiz Future Growth I (already underway). Invest in small/medium onshore KZ fields (Kumkol, North Buzachi) where development costs are lower. In Uzbekistan, target enhanced gas recovery (Shurtan field) and modest oilfield reactivation (Suzaku/Kokdumalak).
- **Medium term (3–7 yr):** Pursue deep expansions: Kashagan Phase 2 (mid-2020s) and Tengiz FGP II. Explore Turkmenistan via PSAs contingent on Trans-Caspian developments: BP could support TAPI or LNG export plans if sanctions ease. In Russia (if sanctions soften), consider equity roles in LNG (Yamal/Arctic) or Far East oil (ESPO–China).
- **Long term (7+ yr):** High-risk frontier: Caspian deepwater (joint with Azeri fields?), Russian Arctic shelf (Vostok Oil), or potential new corridors (Trans-Caspian pipelines). Invest in gas-to-power or petrochemicals domestically to hedge low gas prices. Research shale potential regionally.

These recommendations assume continued global demand and potential easing of political constraints.

Uncertainties and Data Gaps

Key uncertainties include: **Reserve estimates** (especially for Siberian/Ural basins in Russia, and remote Central Asian basins). Official data are often opaque (national statistics vs. international audits).

Prospectivity in undeveloped basins (Arctic Russia, deep Caspian) is largely untested. **Political dynamics** (e.g. future of sanctions, Russia's OPEC+ policy) can rapidly change market access.

Infrastructure project outcomes (e.g. TAPI viability, Trans-Caspian pipeline) remain speculative.

Market conditions (oil price volatility, climate policies) will affect project economics.

Sources: Compiled from government/industry reports and publications [44†L114-L122] [83†L147-L154] [87†L178-L182] . Key references are cited above.
